as a sparse VAR. Since the idiosyncratic component is not observed, its parameters are estimated by an ℓ_1-regularized Yule–Walker estimator built from the idiosyncratic autocovariances implied by the estimated spectral density. The same three networks as in NETS are then estimated, and they reflect only the dependence that cannot be attributed to the common factors. The method also allows the common and idiosyncratic components to be forecast separately. The authors prove the consistency of the estimators as both the number of series and the sample size grow, also under heavier-than-Gaussian tails, and implement the procedure in the R package `fnets` (Owens et al., 2024). For our purpose, FNETS is the key method, because it separates the part of the connectedness among emerging markets that is due to global factors from the part that reflects links between the markets themselves.

1 Motivation

Emerging equity markets are exposed to two kinds of shocks. Some are global: changes in US monetary policy, in the dollar or in investors' appetite for risk move all of them at once, a pattern that Rey (2013) describes as a global financial cycle. Others are local, such as an election, a currency crisis or a change in economic policy in one country, and the question is whether, and through which countries, they spread to the rest. Telling the two apart matters for several reasons. For investors, the benefit of diversifying across emerging markets depends on how much of their co-movement is a common component that no portfolio of these markets can diversify away. For policy makers, a high degree of connectedness calls for different responses if it reflects bilateral transmission (contagion) or a common exposure to the same external shock. For the empirical literature, the distinction is an old one: Forbes and Rigobon (2002) showed that much apparent contagion during crises is high interdependence driven by common shocks and higher volatility.

The tools to measure connectedness have multiplied, and they do not all measure the same thing. The variance-decomposition approach of Diebold and Yılmaz (2014) summarizes how much of each market's forecast-error variance is due to shocks elsewhere; it is dense by construction and does not separate common from bilateral transmission. Sparse network estimators such as NETS (Barigozzi and Brownlees, 2019) keep only the links supported by the data, and FNETS (Barigozzi et al., 2024) first removes the common factor and then estimates the network of what is left. Applied to the same data, these methods can give quite different pictures of how connected markets are, who transmits and who receives, and how stable the network is. Before using a connectedness measure to draw conclusions about contagion, it is therefore worth knowing how much of the measured connectedness depends on the estimation method.

Our sample of 17 emerging markets offers clear examples of both kinds of shock. In the week of the Argentine primary elections of August 2019 the Argentine fund (ARGT) lost 28.4% (in log terms), its largest weekly fall in the sample: a large, country-specific shock. Seven months later, the COVID-19 crash of March 2020 produced the largest weekly losses of almost every market at the same time: 14 of the 17 markets recorded their worst week in March 2020. The Argentine episode illustrates the question we ask: whether a large local shock spreads to other markets, and whether the methods agree on that.

2 Introduction

This paper asks how measured connectedness among emerging equity markets changes with the method used to estimate it, and what share of it reflects a global common factor rather than idiosyncratic links between markets. We study weekly returns of US-listed country funds for 17 emerging markets in Latin America, Asia and EMEA, together with the S&P 500 as a global reference, from March 2011 to October 2026, and we estimate connectedness in four ways: the variance-decomposition measures of Diebold and Yılmaz (2014) on a VAR estimated by OLS;

the same measures on a sparse VAR estimated by the lasso with time-series cross-validation; NETS (Barigozzi and Brownlees, 2019), which estimates sparse Granger, contemporaneous and long-run partial-correlation networks; and FNETS (Barigozzi et al., 2024), which estimates the same networks after removing the common factor. All methods are estimated on the same rolling windows and mapped into comparable measures, and we compare them on three criteria: the density of the networks, the agreement between methods, and their stability over time.

We test three hypotheses. First, regularization alone should matter little for total connectedness at the weekly frequency, because the cross-market dependence of returns is mostly contemporaneous rather than lagged. Second, a substantial share of the connectedness measured by Diebold–Yilmaz should be due to a common factor, so that it falls markedly once the factor is removed, and its time variation should follow the weight of that factor. Third, sparse estimators should produce networks that are less dense, and also less stable from one window to the next, than the dense variance-decomposition network.

The results support the first two hypotheses and only partly the third. The lasso VAR sets almost all lag coefficients to zero yet gives nearly the same connectedness as Diebold–Yilmaz (80.0% against 81.0% on average over windows, correlation 0.98). NETS gives a lower level (61.5%) and keeps about half of the possible long-run links, while tracking the Diebold–Yilmaz series closely (correlation 0.89). FNETS finds a single common factor in every window, which explains on average 58% of the variance of the returns; removing it lowers connectedness by 41% on average, a share that is stable across window lengths, horizons, the exclusion of the S&P 500 and the exclusion of 2020. Over time, Diebold–Yilmaz connectedness and the weight of the common factor move together. The sparse networks are indeed much less dense, but stability depends on the measure: NETS connectedness jumps more from one window to the next than Diebold–Yilmaz, yet its strongest long-run links are more persistent; the idiosyncratic network of FNETS is both the sparsest and the least stable. South Africa, Korea and Taiwan are the main net transmitters and Turkey, Vietnam and Argentina the main net receivers in the dense and NETS systems; once the common factor is removed, Argentina is close to neutral.

Our contribution is to put the base paper's connectedness measures side by side with sparse and factor-adjusted network estimators on the same data, windows and metrics, for a set of emerging markets, and to quantify the part of measured connectedness that is a common factor. Section 3 describes the data, Section 4 the methods and comparison criteria, Section 5 the results, and Sections 6 and 7 discuss them and conclude. Robustness checks are in the appendix.

3 Data

Universe and sample. We use US-listed, USD-denominated country exchange-traded funds (ETFs) as tradable proxies for 17 emerging equity markets: Argentina (ARGT), Chile (ECH), Brazil (EWZ), Mexico (EWW) and Peru (EPU) in Latin America; South Korea (EWY), Taiwan (EWT), India (EPI), China (FXI), Thailand (THD), Indonesia (EIDO), Malaysia (EWM), Vietnam (VNM) and the Philippines (EPHE) in Asia; South Africa (EZA), Turkey (TUR) and Poland (EPOL) in EMEA. The S&P 500 ETF (SPY) is included as a global reference node. Funds were selected if their price history starts by 3 March 2011 (7-day tolerance), at most 5% of weeks are missing, and average daily traded volume is at least USD 3 million. Colombian funds (ICOL, GXG), INDA and MCHI did not pass the criteria; India and China are represented by EPI and FXI. The common sample runs from 11 March 2011 to 23 October 2026, i.e. 816 weeks, with no missing values.

Returns and frequency. Prices are dividend- and split-adjusted, so returns are total returns. We use weekly Friday-to-Friday log returns. Choosing the weekly frequency is a decision of ours, for three reasons: weekly returns retain some cross-market predictability, which is the dynamic dependence a VAR-based network is meant to capture; weekly aggregation mitigates the non-